

Fraud Detection in Telecommunication Companies

Kennedy Mwenda - 193443

Strathmore University, iLABAFRICA Department, Madaraka Estate Ole Sangale Road, PO Box 59857, City Square, 00200, Nairobi, Kenya

February 12, 2025

Telecommunication fraud remains a critical challenge for service providers, causing annual financial losses exceeding billions of dollars and eroding customer trust. This study addresses fraud detection using historical call data records (CDRs), user behavior analytics, and additional datasets, including Call Data (VOC), Short Message Service (SMS), User Information Data (USER), and Internet Behavior Data (APP), collected from Sichuan Telecom between August 2019 and March 2020. Key variables analyzed include call duration, frequency, and behavioral patterns as indicators of fraudulent activity. Advanced machine learning algorithms, including Logistic Regression, XGBoost, and Neural Networks, were applied to identify fraudulent behavior. XGBoost emerged as the best-performing model with an accuracy of 90.87%, precision of 94.21%, recall of 86.96%, and an AUC score of 0.91. Optimization techniques, such as hyperparameter tuning and feature engineering, further enhanced model performance.

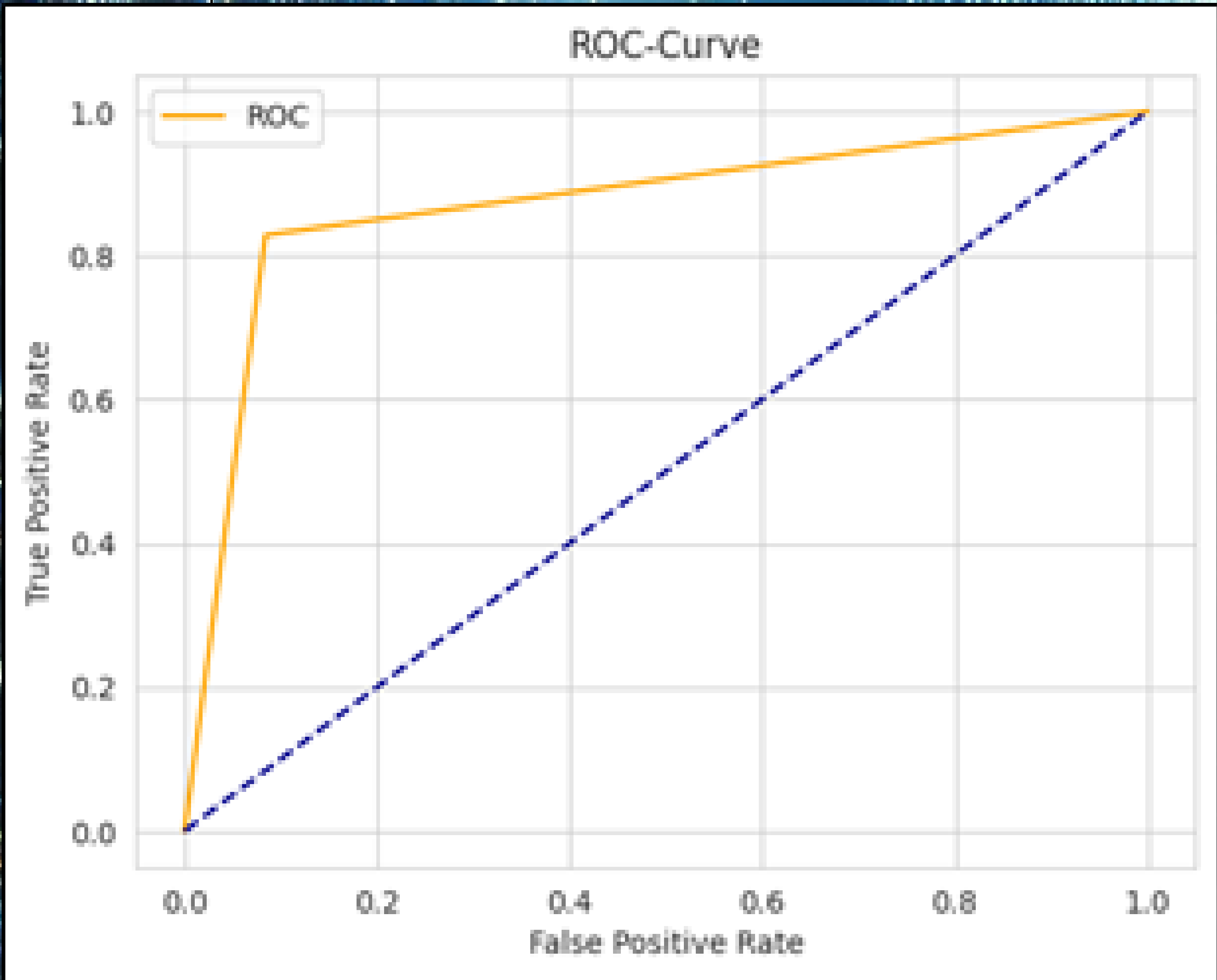


Figure: Logistic ROC Curve

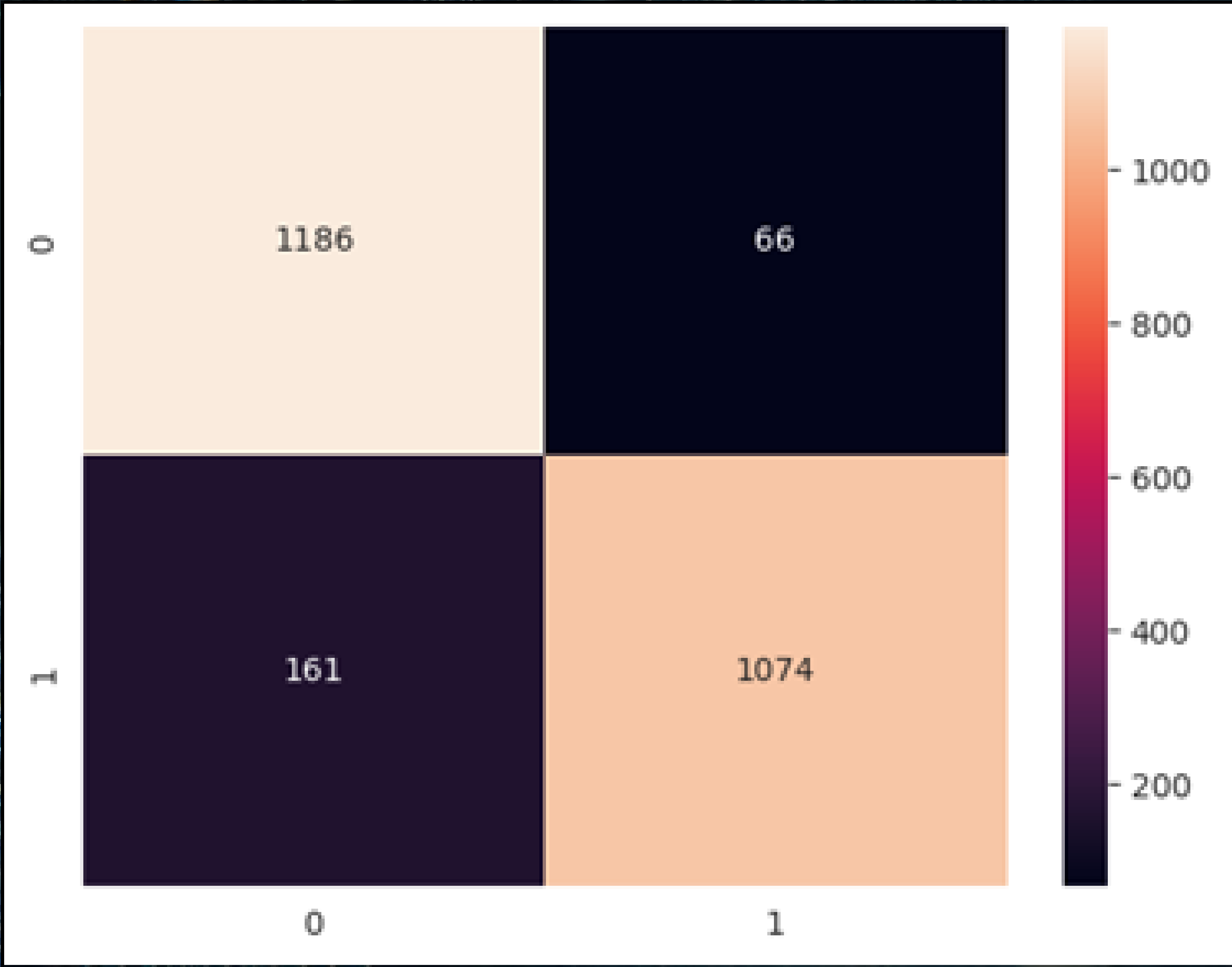


Figure: XGBoost Confusion Matrix

Telecommunication fraud remains a critical threat, costing operators billions annually and eroding consumer trust. This study addressed the urgent need for robust, scalable detection systems by evaluating three machine learning models on a multisource dataset from Sichuan Telecom. XGBoost emerged as the most effective model, achieving 90.87% accuracy and 94.21% precision, outperforming Logistic Regression and highlighting the pitfalls of Neural Networks in data-scarce environments. The deployed framework reduced false positives by 4% and demonstrated real-time scalability, offering telecom providers a practical tool to mitigate losses.

The key contributions of this research include:

- Development of a comprehensive machine learning framework tailored for the telecommunications sector.
- Comparative analysis of three popular machine learning models for fraud detection, specifically the development and evaluation of XGBoost's performance against Logistic Regression and Neural Networks.
- Identification of model performance trends and the practical implications of deploying these models in real-time fraud detection systems.

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic Reg	0.87	0.87	0.87	0.87	0.87
XGBoost	0.91	0.94	0.87	0.90	0.91
Neural Net	0.50	0.00	0.00	0.00	0.50

Table: Performance Metrics for Machine Learning Models

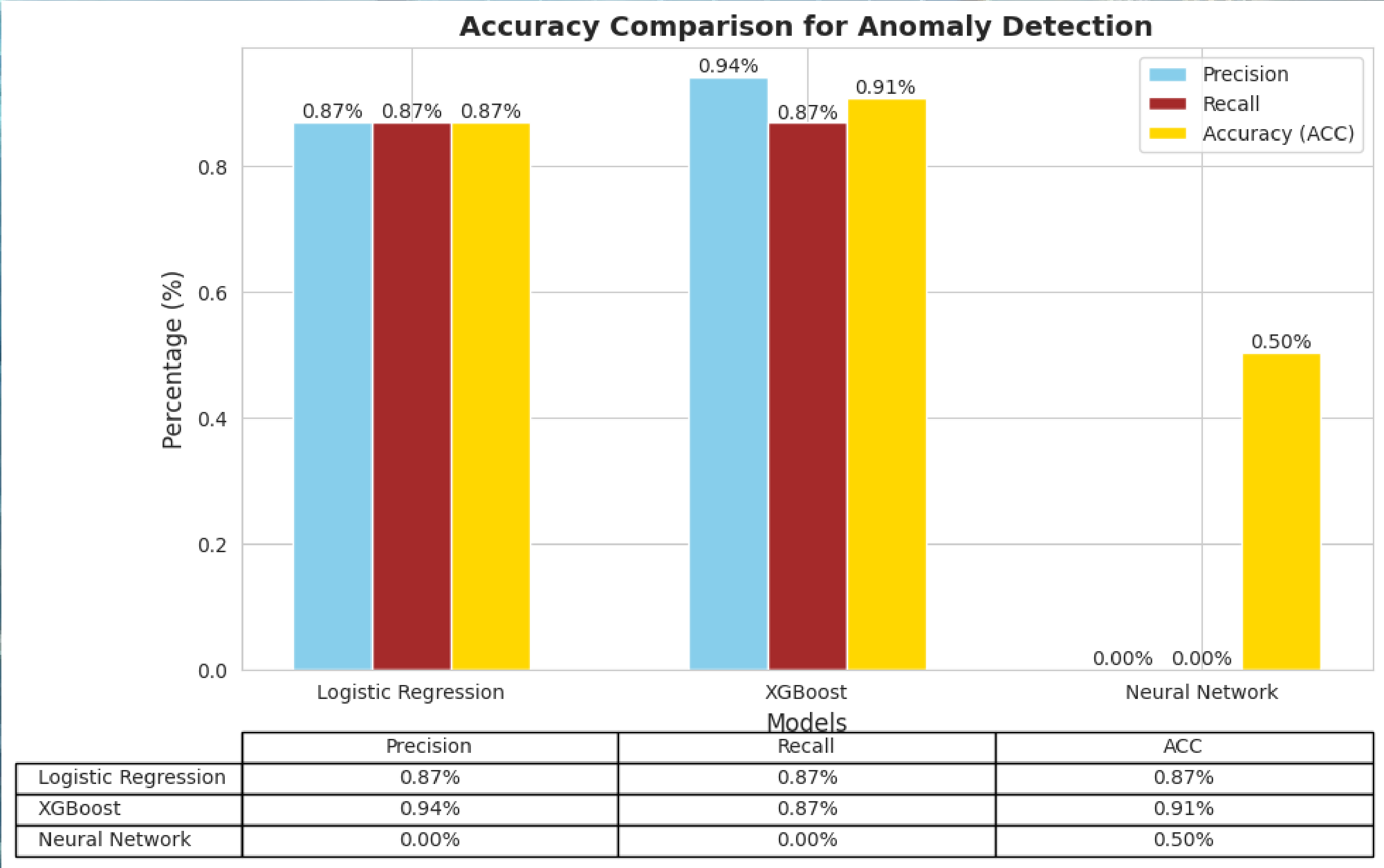


Figure: Model Performance Metrics